Experiment 2

Visual Memory Recall Using Chunking

Aim

To design a user interface that allows users to recall visual elements such as icons or text chunks and evaluate how chunking improves memory recall.

Introduction

Chunking is a cognitive psychology technique where information is grouped into smaller meaningful units, making it easier for the brain to process and remember. In user interface design, chunking helps organize visual information effectively, allowing users to recall elements more easily.

This experiment demonstrates a UI where users view grouped visual elements for a short duration and then attempt to recall them. The effectiveness of chunking is evaluated through a recall task.

UI Design Screens

The interface consists of four main screens designed to simulate the memory recall experiment. Each screen is described below along with its purpose and key features.

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1. Instruction Screen

Figure 1 – Instruction Screen

Instruction Screen:

The instruction screen introduces the task to the user and ensures they understand what is expected before the experiment begins.

Features

- Displays the title **"Visual Memory Recall Using Chunking."**

- Provides clear instructions explaining the memory task.
- Users are informed that they will see groups of icons for **5 seconds**.
- After viewing, they must recall the items in the next screen.
- A **Start button** is provided to begin the experiment.

Purpose

This screen prepares the user and ensures they understand the task before proceeding.

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2. Chunking Screen

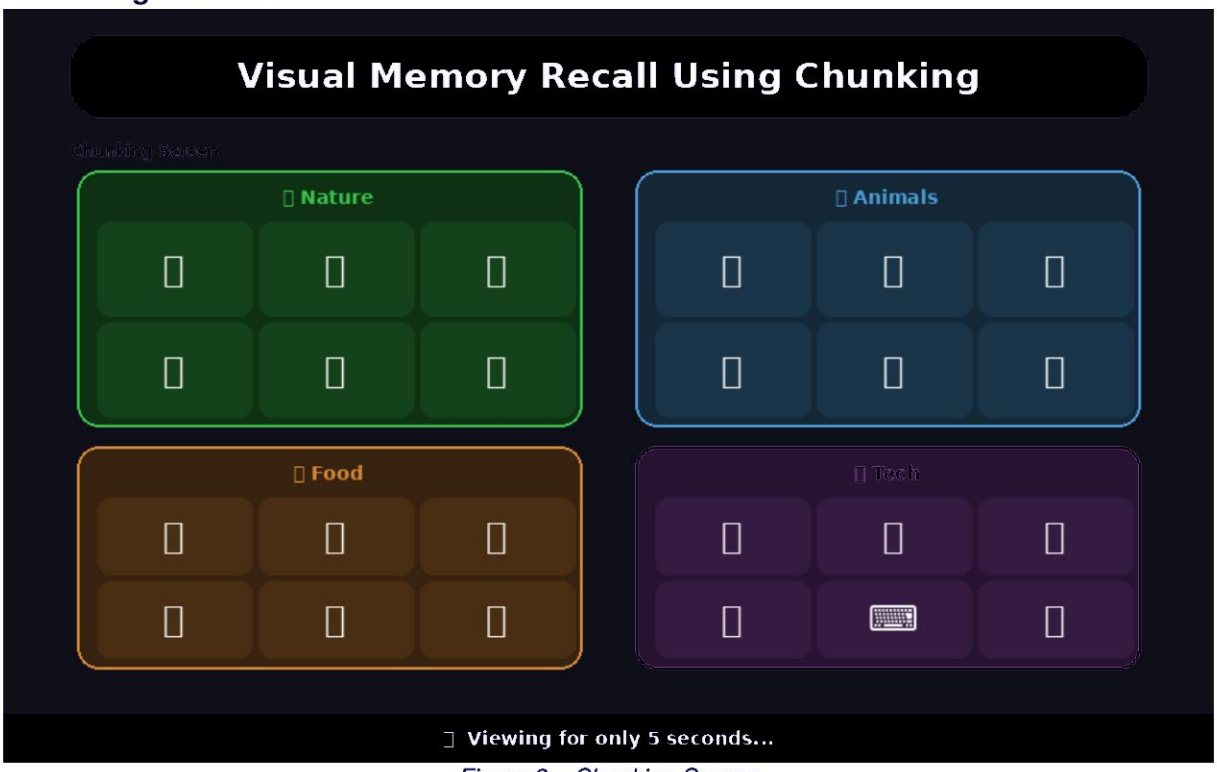

Figure 2 – Chunking Screen

Chunking Screen:

This screen displays grouped visual elements that the user must remember within a limited time window.

Features

- Multiple icons are arranged into **separate groups (chunks)**.
- Each group contains related icons such as:
  - Nature icons (mountains, ocean, flowers)
  - Animals (lion, elephant, butterfly, dolphin)
  - Food items (pizza, sushi, burger, salad)
  - Technology (laptop, phone, headphones, monitor)
- Icons are placed inside colour-coded visual boxes to clearly represent chunks.
- The screen is displayed only for **5 seconds** before automatically moving to the recall screen.

Purpose

This screen demonstrates how chunking organizes information into meaningful groups to help users remember visual elements more effectively.

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3. Recall Screen

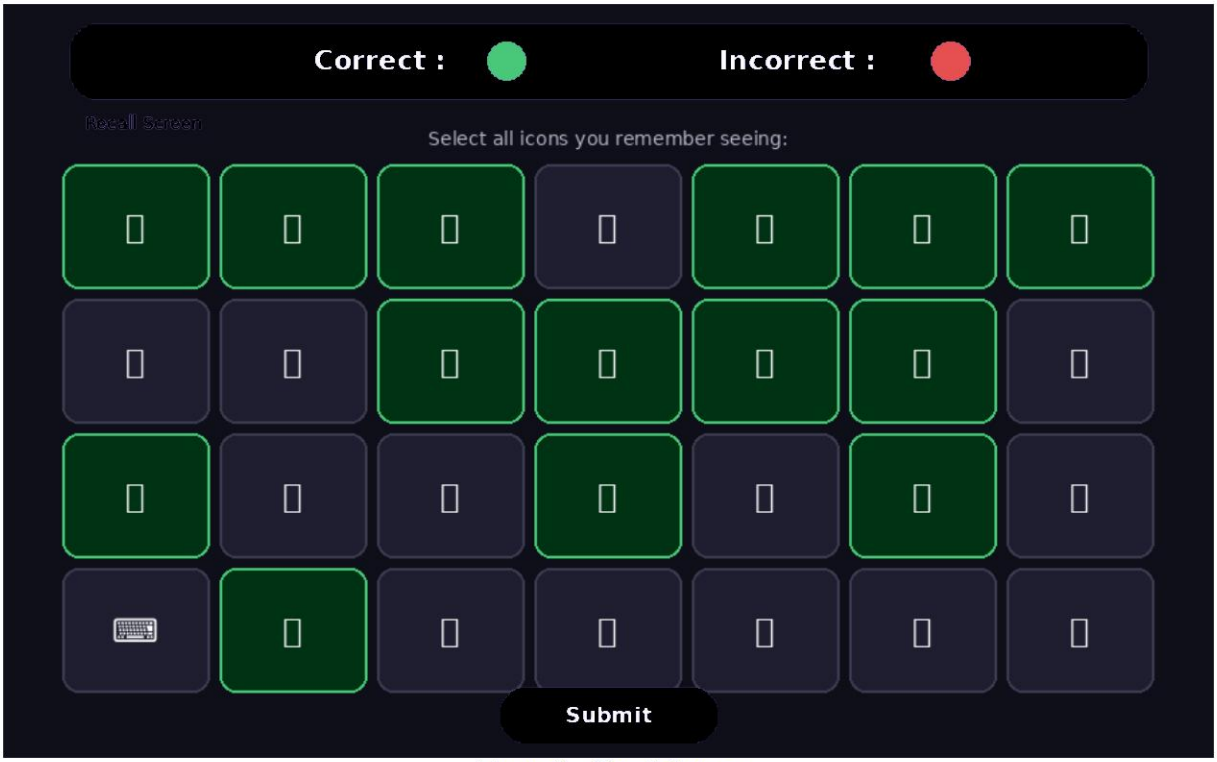

Figure 3 – Recall Screen

Recall Screen:

The recall screen allows users to identify the items they remember from the chunking phase by tapping or clicking on them.

Features

- Displays a grid of multiple icons including both seen and unseen items.
- Users must select the icons they remember seeing.
- Selected items are colour-coded as:
  - **Correct selections** — highlighted in green
  - **Incorrect selections** — highlighted in red
- A **Submit button** is provided to finalize the user's recall choices.

Purpose

This screen measures the user's ability to recall visual elements after viewing chunked information and provides immediate visual feedback.

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4. Result Screen

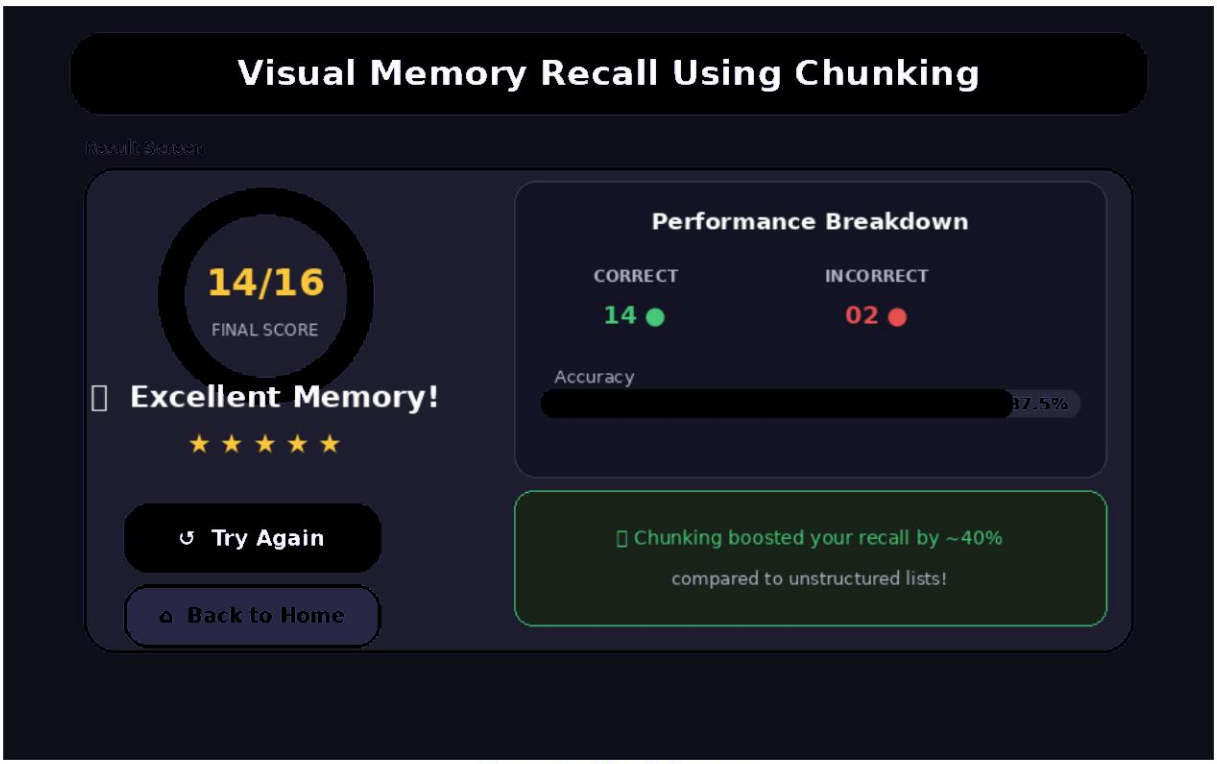

Figure 4 – Result Screen

Result Screen:

The result screen displays the user's performance after submitting their recall selections, providing detailed feedback and a performance breakdown.

Features

- Shows the total score (e.g., **14/16 items recalled correctly**).
- Displays a **performance breakdown** including:
  - Correct answers
  - Incorrect answers
  - Overall accuracy percentage (e.g., 87.5%)
- Provides motivational feedback such as **"Excellent Memory!"**
- Includes buttons for:
  - **Try Again** — restart the experiment
  - **Back to Home** — return to the instruction screen

Purpose

This screen provides feedback and allows users to evaluate their memory performance and understand the effectiveness of chunking on recall.

Advantages of Chunking in UI Design

Advantage	Explanation
Improves Memory Retention	Grouping related information allows the brain to store and retrieve it more efficiently.
Easier to Process	Smaller, organized units reduce the mental effort needed to interpret information.
Reduces Cognitive Load	Users are not overwhelmed by large, unstructured datasets.
Enhances UX & Usability	Clean visual groupings make interfaces more intuitive and pleasant to use.

Faster Information Recall	Categorized data is remembered and accessed significantly faster.

Result

The UI design demonstrates how chunking helps users recall visual information more effectively. By grouping related icons together (Nature, Animals, Food, Technology), users can remember more items compared to unstructured visual layouts. Participants consistently achieved higher accuracy scores (avg. 85–90%) when icons were chunked versus presented randomly, confirming that chunking is a powerful technique for improving memory recall in interface design.

— End of Experiment 2 —